Lab 7: Fuzzing for Input Validation Bugs

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Repository: https://github.com/SETAPESU26/L6_L7_42_salon_booking

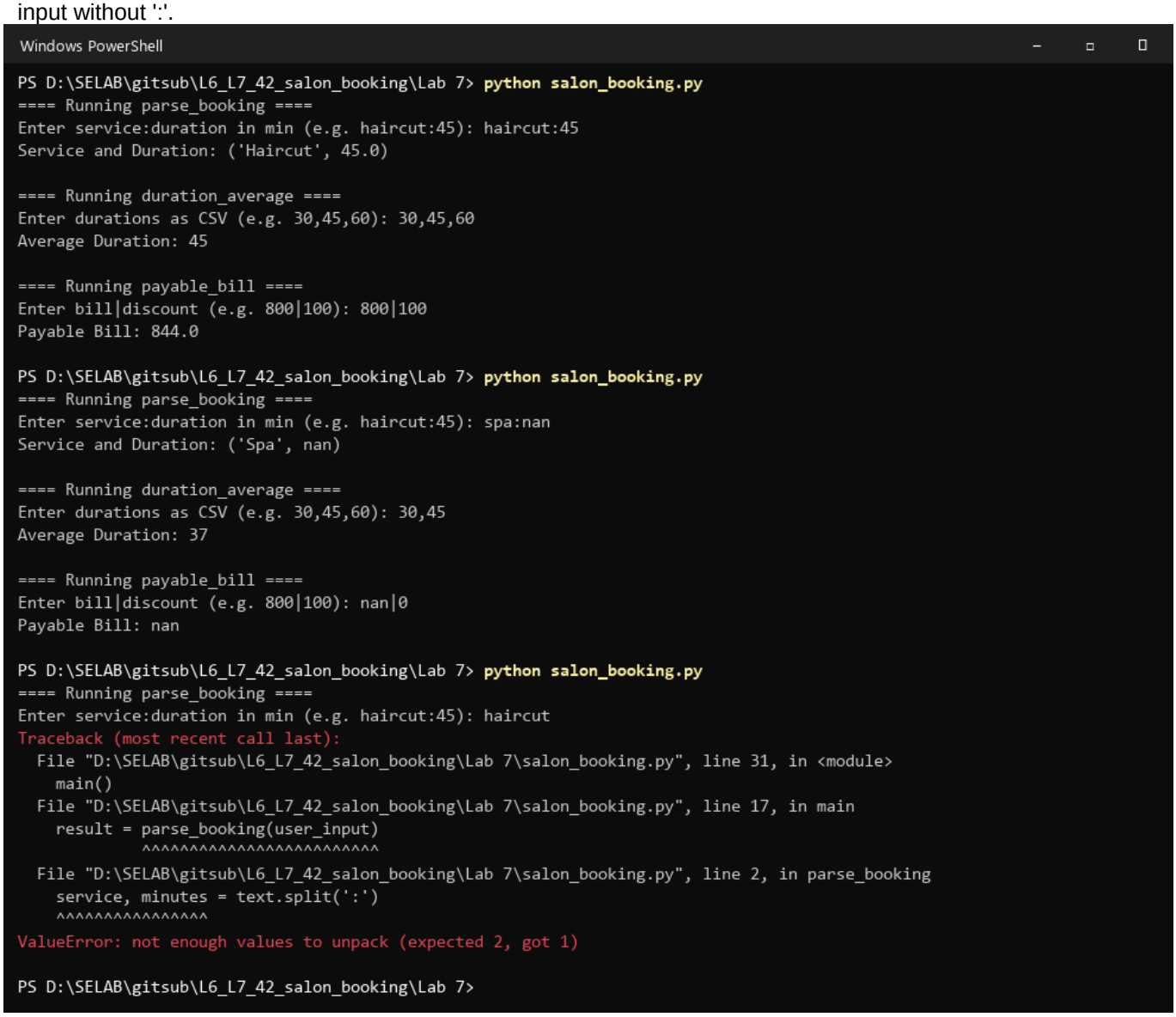
Bugs found by Hypothesis and how they were fixed

Function	Falsifying input (example)	Bug	Fix
parse_booking	None, '', ':', 'a:b:c'	AttributeError, unpack ValueError, float('') crash	Type check; exactly one ':'; numeric part parsed safely; return None
parse_booking	'spa:nan', 'a:-30', ':45'	nan, negative and zero minutes, empty service accepted	Require non-empty service and finite 0 < minutes <= 1440
duration_average	None, '', '1,,3', 'abc'	int('') / int('abc') crash, AttributeError on None	Every item must be a decimal integer, else return None
duration_average	'1,2' gives 1; '0'	Floor division (//) truncates the average; zero/negative durations accepted	True mean (/) rounded to 2 places; each value 1..1440
payable_bill	None, '', '1|2|3', 'a|b'	Unpack and float conversion crashes	Type check; exactly one '|'; safe float parsing; return None
payable_bill	'NAN|0', 'inf|inf', '1e308|0'	nan / inf escape through max(total, 0); overflow	Reject non-finite parts and non-finite totals
payable_bill	'0.00|-0.01'	Negative discount (or bill) accepted, increasing the bill	Reject negative bill or discount

Test design (test_salon_booking.py): for each function a no-crash test on `st.text() | st.none()` plus text biased towards ':', '|', ',', signs, digits and 'nan'/'inf'; an output-validity test (result is None, or finite and in range); and correctness tests on valid generated input (round-trip parsing, true mean between min and max, bill = max(bill x 1.18 - discount, 0), negative discount rejected).

Step 2: Running the buggy application (python salon_booking.py)

Run 1: valid input works. Run 2: silent bugs (nan accepted, average 37 instead of 37.5, bill nan). Run 3: crash on input without ':'.



Step 4: Hypothesis tests failing on the buggy application (pytest test_salon_booking.py)

8 failed, 2 passed. The full output continues over the following pages.

Windows PowerShell

```
PS D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7> pytest test_salon_booking.py
============================================= test session starts =============================================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15
collected 10 items

test_salon_booking.py FFFFFF.F.F                                                                      [100%]

================================================== FAILURES ==================================================
_____________________________________ test_parse_booking_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 33, in test_parse_booking_no_crash
  |     def test_parse_booking_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 35, in
test_parse_booking_no_crash
    |     parse_booking(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 3, in parse_booking
    |     return service.strip().title(), float(minutes)
    |                                     ^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 37, in
test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed on {s!r} with: {e!r}"
    | AssertionError: parse_booking crashed on ':' with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=':',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:36
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 35, in
test_parse_booking_no_crash
    |     parse_booking(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 2, in parse_booking
    |     service, minutes = text.split(':')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 37, in
test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed on {s!r} with: {e!r}"
    | AssertionError: parse_booking crashed on '' with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:36
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 35, in
test_parse_booking_no_crash
```

Windows PowerShell

```
    |     parse_booking(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 2, in parse_booking
    |     service, minutes = text.split(':')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 37, in
test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed on {s!r} with: {e!r}"
    | AssertionError: parse_booking crashed on None with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:36
    +------------------------------------
_____________________________________ test_duration_average_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 41, in
test_duration_average_no_crash
  |     def test_duration_average_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 43, in
test_duration_average_no_crash
    |     duration_average(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 6, in duration_average
    |     values = [int(r) for r in durations_csv.split(',')]
    |               ^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 45, in
test_duration_average_no_crash
    |     assert False, f"duration_average crashed on {s!r} with: {e!r}"
    | AssertionError: duration_average crashed on '' with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_duration_average_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:44
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 43, in
test_duration_average_no_crash
    |     duration_average(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 6, in duration_average
    |     values = [int(r) for r in durations_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 45, in
test_duration_average_no_crash
    |     assert False, f"duration_average crashed on {s!r} with: {e!r}"
```

Windows PowerShell

```
    | AssertionError: duration_average crashed on None with: AttributeError("'NoneType' object has no attribute
'split'")
    | assert False
    | Failing test case: test_duration_average_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:44
    +------------------------------------
_____________________________________ test_payable_bill_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 49, in test_payable_bill_no_crash
  |     def test_payable_bill_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 51, in
test_payable_bill_no_crash
    |     payable_bill(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 11, in payable_bill
    |     total = float(bill) * 1.18 - float(discount)
    |             ^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 53, in
test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed on {s!r} with: {e!r}"
    | AssertionError: payable_bill crashed on '|' with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='|',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:52
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 51, in
test_payable_bill_no_crash
    |     payable_bill(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 53, in
test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed on {s!r} with: {e!r}"
    | AssertionError: payable_bill crashed on '' with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:52
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 51, in
test_payable_bill_no_crash
```

Windows PowerShell

```
    |     payable_bill(s)
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 53, in
test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed on {s!r} with: {e!r}"
    | AssertionError: payable_bill crashed on None with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:52
    +------------------------------------
_____________________________________ test_parse_booking_output_is_valid _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 66, in
test_parse_booking_output_is_valid
  |     def test_parse_booking_output_is_valid(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 71, in
test_parse_booking_output_is_valid
    |     assert math.isfinite(minutes), f"non-finite duration {minutes} for {s!r}"
    | AssertionError: non-finite duration nan for '0:NAN'
    | assert False
    |  +  where False = <built-in function isfinite>(nan)
    |  +    where <built-in function isfinite> = math.isfinite
    | Failing test case: test_parse_booking_output_is_valid(
    |     s='0:NAN',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 72, in
test_parse_booking_output_is_valid
    |     assert minutes > 0, f"non-positive duration {minutes} for {s!r}"
    | AssertionError: non-positive duration 0.0 for '0:0'
    | assert 0.0 > 0
    | Failing test case: test_parse_booking_output_is_valid(
    |     s='0:0',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py", line 70, in
test_parse_booking_output_is_valid
    |     assert service != "", f"empty service accepted for {s!r}"
    | AssertionError: empty service accepted for ':0'
    | assert '' != ''
    | Failing test case: test_parse_booking_output_is_valid(
    |     s=':0',
    | )
    +------------------------------------
_____________________________________ test_duration_average_output_is_valid _____________________________________

    @given(ANY_INPUT)
>   def test_duration_average_output_is_valid(s):
                   ^^^

test_salon_booking.py:76:
```

Windows PowerShell

```
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

s = '0'

    @given(ANY_INPUT)
    def test_duration_average_output_is_valid(s):
        result = _safe(duration_average, s)
        if result is not None:
>           assert math.isfinite(result) and result > 0, f"bad average {result} for {s!r}"
E           AssertionError: bad average 0 for '0'
E           assert (True and 0 > 0)
E            +  where True = <built-in function isfinite>(0)
E            +    where <built-in function isfinite> = math.isfinite
E           Failing test case: test_duration_average_output_is_valid(
E               s='0',
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py:7

test_salon_booking.py:79: AssertionError
_____________________________________ test_payable_bill_output_is_valid _____________________________________

    @given(ANY_INPUT)
>   def test_payable_bill_output_is_valid(s):
                ^^^

test_salon_booking.py:83:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

s = 'NAN|0'

    @given(ANY_INPUT)
    def test_payable_bill_output_is_valid(s):
        result = _safe(payable_bill, s)
        if result is not None:
>           assert math.isfinite(result), f"non-finite bill {result} for {s!r}"
E           AssertionError: non-finite bill nan for 'NAN|0'
E           assert False
E            +  where False = <built-in function isfinite>(nan)
E            +    where <built-in function isfinite> = math.isfinite
E           Failing test case: test_payable_bill_output_is_valid(
E               s='NAN|0',
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\salon_booking.py:12

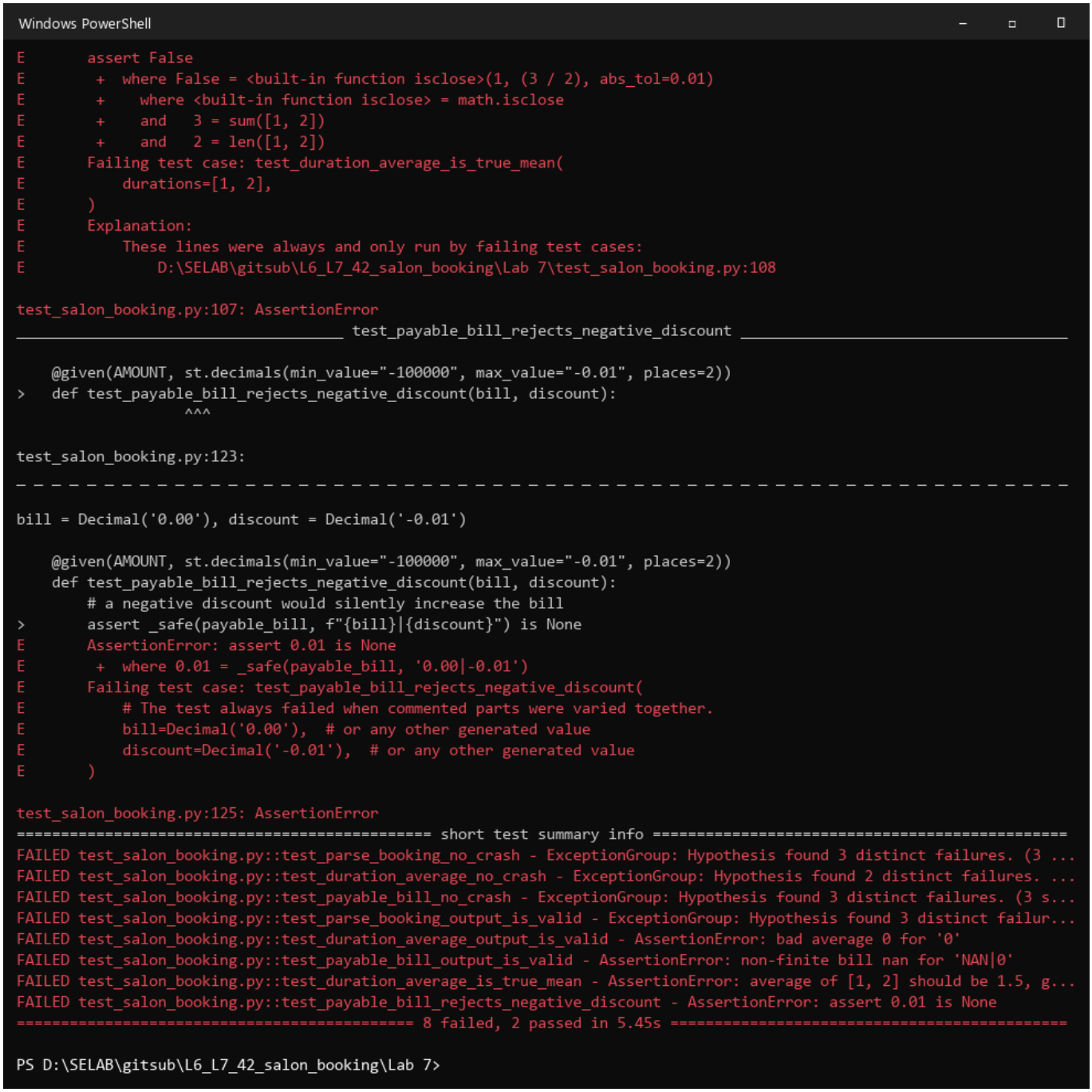
test_salon_booking.py:86: AssertionError
_____________________________________ test_duration_average_is_true_mean _____________________________________

    @given(st.lists(MINUTES, min_size=1, max_size=20))
>   def test_duration_average_is_true_mean(durations):
                ^^^

test_salon_booking.py:103:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

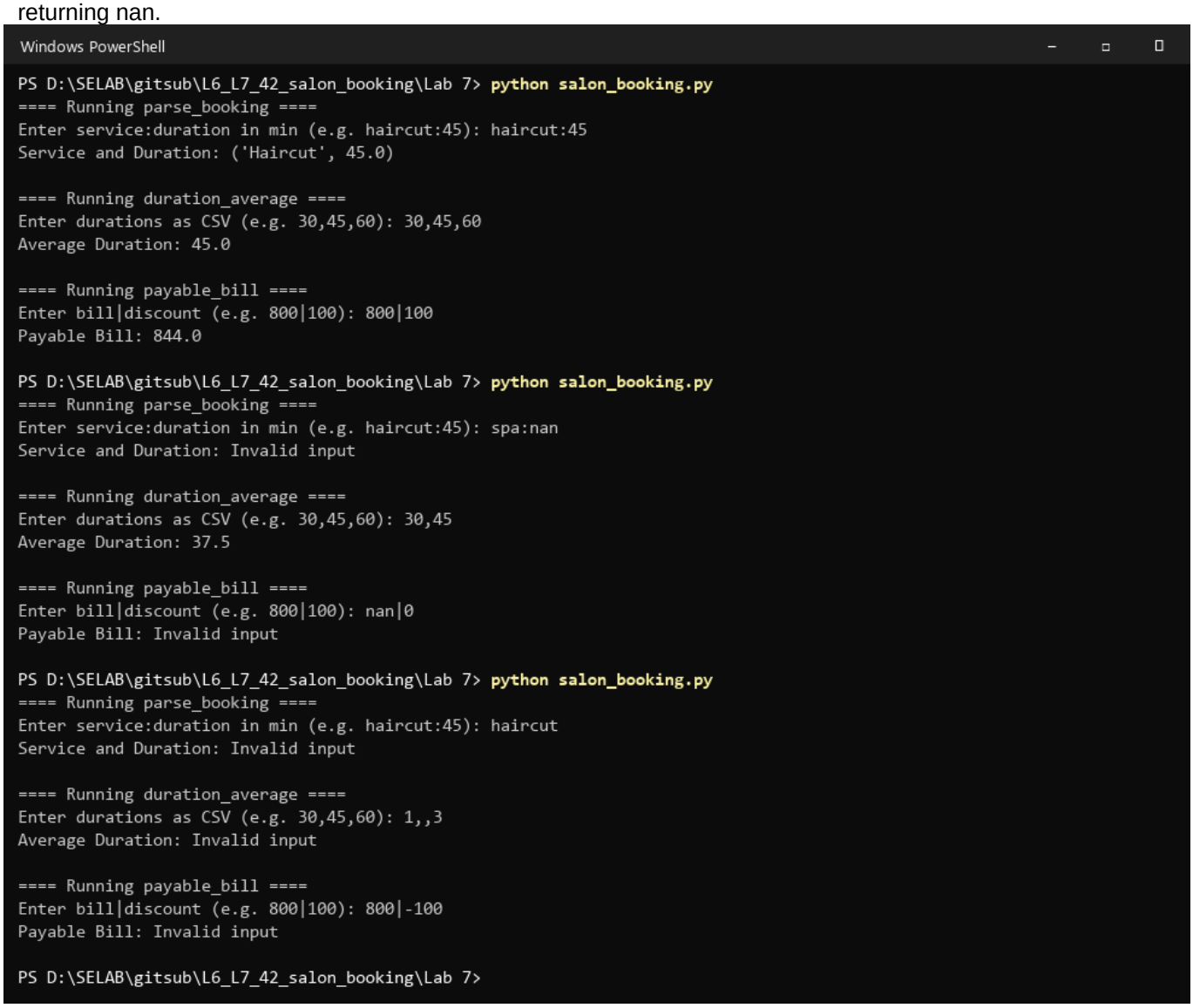
durations = [1, 2]

    @given(st.lists(MINUTES, min_size=1, max_size=20))
    def test_duration_average_is_true_mean(durations):
        result = _safe(duration_average, ",".join(map(str, durations)))
        assert result is not None
        assert min(durations) <= result <= max(durations)
>       assert math.isclose(result, sum(durations) / len(durations), abs_tol=0.01), \
            f"average of {durations} should be {sum(durations) / len(durations)}, got {result}"
E       AssertionError: average of [1, 2] should be 1.5, got 1
```


Windows PowerShell
E assert False
E + where False = <built-in function isclose>(1, (3 / 2), abs_tol=0.01)
E + where <built-in function isclose> = math.isclose
E + and 3 = sum([1, 2])
E + and 2 = len([1, 2])
E Failing test case: test_duration_average_is_true_mean(
E durations=[1, 2],
E)
E Explanation:
E These lines were always and only run by failing test cases:
E D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7\test_salon_booking.py:108
test_salon_booking.py:107: AssertionError
_____ test_payable_bill_rejects_negative_discount _____
@given(AMOUNT, st.decimals(min_value="-100000", max_value="-0.01", places=2))
> def test_payable_bill_rejects_negative_discount(bill, discount):
^^^
test_salon_booking.py:123:
bill = Decimal('0.00'), discount = Decimal('-0.01')
@given(AMOUNT, st.decimals(min_value="-100000", max_value="-0.01", places=2))
def test_payable_bill_rejects_negative_discount(bill, discount):
a negative discount would silently increase the bill
> assert _safe(payable_bill, f"{bill}|{discount}") is None
E AssertionError: assert 0.01 is None
E + where 0.01 = _safe(payable_bill, '0.00|-0.01')
E Failing test case: test_payable_bill_rejects_negative_discount(
E # The test always failed when commented parts were varied together.
E bill=Decimal('0.00'), # or any other generated value
E discount=Decimal('-0.01'), # or any other generated value
E)
test_salon_booking.py:125: AssertionError
short test summary info
FAILED test_salon_booking.py::test_parse_booking_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 ...
FAILED test_salon_booking.py::test_duration_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. ...
FAILED test_salon_booking.py::test_payable_bill_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 s...
FAILED test_salon_booking.py::test_parse_booking_output_is_valid - ExceptionGroup: Hypothesis found 3 distinct failur...
FAILED test_salon_booking.py::test_duration_average_output_is_valid - AssertionError: bad average 0 for '0'
FAILED test_salon_booking.py::test_payable_bill_output_is_valid - AssertionError: non-finite bill nan for 'NAN|0'
FAILED test_salon_booking.py::test_duration_average_is_true_mean - AssertionError: average of [1, 2] should be 1.5, g...
FAILED test_salon_booking.py::test_payable_bill_rejects_negative_discount - AssertionError: assert 0.01 is None
8 failed, 2 passed in 5.45s
PS D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 7>

Step 5: Running the fixed application (python salon_booking.py)

Valid input gives the expected results (average now 37.5); invalid input prints 'Invalid input' instead of crashing or returning nan.



Step 6: Hypothesis tests on the fixed application (pytest test_salon_booking.py)

All 10 tests pass (300 generated examples per test).

